

CS465 Research Project: VR Place Tag

Evaluating Gesture-Based Interaction for Placing and Labeling Spatial Markers in Augmented Reality

Group Name: VR Place Tag

Dionicio Flores (Implementation Lead)

Daniel Steven (Study Design Lead)

Matt Li (Data Analysis Lead)

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1 Abstract

Augmented reality (AR) systems increasingly rely on natural interaction methods such as gesture and voice input to allow users to interact with digital content embedded in physical spaces. This project explores how users place and label spatial markers in AR environments using gesture interaction techniques. The goal of the study was to evaluate the usability and user satisfaction of a hands-free AR labeling system designed for shared and dynamic environments.

A prototype AR application was developed that allows users to place spatial markers using hand gestures and assign predefined labels to those locations. A within-subjects user study involving 12 participants was conducted to measure task completion time, error frequency, false tag placement (incorrectly placed labels), and user satisfaction. Participants also completed a five-question interview covering usability, intuitiveness, and prior AR/VR experience.

Results showed that users with prior AR/VR experience generally completed tasks faster and with fewer errors, while new users required additional time to learn the interaction system. Although several participants experienced accidental tag placements and gesture recognition issues, most participants reported positive satisfaction scores and described the interaction as intuitive after a short adjustment period. These findings suggest that gesture-based AR labeling systems can effectively support spatial memory and navigation tasks, though improvements in gesture precision and feedback mechanisms are needed to reduce user errors and improve accessibility for novice users.

2 Introduction

Augmented reality (AR) can help people keep track of important locations in the physical world: At events, conferences, workshops, or marketplaces, it is easy to lose track of where your table is, where your group is sitting, where equipment is stored, or even where you parked your car. People often rely on memory or temporary visual cues, but once a space fills with people, those cues disappear. This project explores how AR can support these everyday situations by allowing users to level a place and later retrieve spatial markers tied to meaningful locations. As users move through a space, they can drop AR markers on places that matter to them, such as a table, meeting point, group location, exit, or practical spots like a safety gear. These markers can be viewed later by the same user or shared with others. Marker placement is designed to feel simple: a natural hand gesture, such as pinching and releasing, places a marker at the location the user is looking at. This avoids a controllers and keeps the interaction intuitive. The main research challenge arises when labeling these markers. So hand gestures are the most straight forward solutions, so we want to figure out what combinations and interaction well be the most practical.

Gesture-based labeling offers an alternative that is unaffected by noise and allows users to select from predefined icons or labels using simple actions such as pinching. Although gestures are reliable, they limit expressiveness and prevent users from creating fully personalized labels. So far, our contributions to these past studies include building a software that can label places. We conducted a research to see how effective gestures are truly. This project presents an AR system for placing and labeling spatial tags. By comparing speed, reliability, error rate, and user preference, this work aims to better understand input modalities.

3 Related Work

We based our experiment on 3 key AR research papers, and other papers also support our findings. . Mark Billinghurst [2] established that labeling and annotating physical spaces is a “core AR task.” This shows that our project is addressing

a problem to how we attach digital information to the real world rather than just a technical issue. To focus more specifically on input methods, we used the 2024 study by Korkiakoski [1] as a baseline for understanding gesture-based interaction in AR systems. Their research found that gesture controls can feel intuitive and reliable for novice users after a short learning period. These findings helped motivate our evaluation of gesture-based spatial labeling and user satisfaction within our own AR prototype. Finally, we incorporate research from Qiu[3] on cognitive load. Their work highlights that poorly designed AR interfaces can distract users and reduce their awareness of their surroundings. This supports our decision to go beyond just measuring task time with system logs and also collect user feedback on frustration. For example, if a user has to repeat a voice command multiple times for it to work, their mental effort is being spent on their ability rather than the task itself. Measuring that extra cognitive load is a key part of what we aim to understand and improve.

Zhang et al. [13] ran an elicitation study to understand how users naturally perform mid-air gestures in AR. They found that participants preferred simple, one handed gestures and that gesture choice varied with prior AR/VR experience. This aligns with our observation that participants with previous VR exposure adapted more quickly and made fewer errors, while novice users required more time to learn the gesture vocabulary. Together, these findings support the idea that gesture based systems must accommodate varying experience levels to be more usable.

Mid-air gestures feel natural to use, but they don't always work perfectly. Nguyen [15] studied different hand gestures used in AR headsets and found that even simple actions, like pinching, can be misunderstood by the system. When the system misreads a gesture, it can trigger actions the user didn't intend. This matches what we saw in our study: participants sometimes placed tags by accident or had to repeat a gesture before the system recognized it. To fully understand these problems, we looked at more than just how fast people finished the task. Qiu [3] explained that when an AR interface is hard to use, it increases mental effort and takes attention away from what the user is actually trying to do. Following this idea, we measured both task performance time and errors, and user experience (satisfaction and frustration). This gave us a more complete view of how the gesture system affected users.

3.1 AR Labeling and Interaction

Research in AR labeling and annotation shows that attaching digital information to physical locations is a core AR task. Billingham et al. provide a foundational survey of AR systems and highlight challenges such as visual clutter, occlusion, spatial accuracy, and interaction efficiency [2]. Their work demonstrates that labeling is both common and technically demanding, making it an ideal task for evaluating input methods.

3.2 Voice vs. Gesture Input

Korkiakoski et al. studied how novice users interact with hands-free AR systems using both voice commands and gesture controls [1]. They found that voice input is efficient in quiet environments but becomes unreliable as noise increases, while gesture input remains stable but sometimes difficult for new users. These findings highlight a key trade-off between expressiveness and reliability, and motivate the need to evaluate both input methods under varying environmental conditions.

3.3 Cognitive and User Experience Factors

Recent work has emphasized the importance of cognitive and experiential factors in AR interactions. Qiu et al. [3] reviewed a range of AR wayfinding studies and found that while AR can improve task performance and situational awareness, user experience and cognitive outcomes are highly dependent on interface design and immersion. Their findings highlight that usability, cognitive load, and user satisfaction are critical components of AR evaluation. Additionally, inconsistencies in experimental design across prior studies leave open questions about the effectiveness of different interaction modalities. These results suggest that evaluating both performance and user experience is essential when comparing input methods in AR environments.

4 Reserach Question

How do voice and gesture input methods compare in terms of speed, error rate, and user preference when labeling spatial markers in AR?

For or hypothesis we predetermined that participants with prior AR/VR experience will complete tasks faster, make fewer errors, and report higher satisfaction. Also, the most common errors will be accidental tag placements caused by gesture misinterpretation.

5 Methodology

5.1 Participants

The study involved 12 participants recruited from Fort Collins, either current students or alumni of Colorado State University. Prior AR/VR experience varied across participants, with only two participants reporting previous experience using VR systems. Most participants were right-hand dominant, while three participants identified as left-handed.

Participants volunteered to participate in the study and were not financially compensated. Before beginning the experiment, each participant was briefly introduced to the AR system and informed about the study procedure. Demographic information collected during the interview included participant name, age, prior AR/VR experience, and hand preference.

5.2 Apparatus

The experiment used a custom-built augmented reality application called AR Place Tag. The project was developed to allow users to place spatial markers within a physical environment using hand gestures. Marker placement was triggered through pinching gestures, while marker labeling used a predefined icon and menu selections.

The application was designed to support hands-free interaction without requiring external controllers. Participants interacted with the system using natural hand gestures while wearing an AR/VR headset capable of hand tracking and spatial anchoring. The experiment recorded task completion time, placement errors, and false labeling during each trial.

The interface included:

- Gesture-based marker placement
- Predefined spatial labels/icons
- Persistent AR tags anchored in physical space
- Visual feedback confirming successful placement
- Spatial marker retrieval and viewing

5.3 Procedure

Each participant completed an individual testing session using the AR Place Tag application. At the start of the session, participants were introduced to the AR system and shown how to perform the required hand gestures for placing and labeling spatial markers.

Participants first completed a short demonstration and practice session to familiarize themselves with the interaction system. After practice, participants performed a series of spatial tagging tasks in the AR environment. All testing sessions were conducted in a quiet indoor environment with minimal background noise or external distractions. This controlled setting helped reduce the amount of potentially distracting factors that could interfere with gesture tracking or participant concentration. The primary task required participants to place markers at specified physical locations and assign labels using the gesture-based interface.

Timing began when participants were told to start the marker placement test and ended once all required markers had been successfully placed and labeled. The data collected from the experiment includes:

- Total task completion time
- Number of placement or interaction errors
- Number of false labels accidentally created
- Participant satisfaction ratings
- Observational usability notes

Errors included incorrect gesture recognition, unintended activations, or failed placement attempts. False tags referred to markers accidentally placed at unintended locations. Participant satisfaction was measured using a self reported Likert style scale ranging from 1 to 10.

After completing the task, participants participated in a short interview. The interview collected qualitative feedback regarding usability, intuitiveness, comfort, and potential improvements for the system. The interview questions included:

1. What is your name, age, gender, and major?

2. Do you have any experience with AR/VR?
3. Can you explain how it felt to use our application?
4. Any further explanations?
5. What feedback do you have about our product? Something to change to make it feel better and more intuitive to use?

6 Results

6.1 Quantitative Results

Twelve participants completed the AR spatial labeling task using the AR Place Tag application. Across all participants, the average task completion time was approximately 4 minutes and 2 seconds. Completion times ranged from 1 minute and 37 seconds to 7 minutes and 3 seconds.

The average number of interaction errors per participant was approximately 1.9 errors, while the average number of false tag placements was 2.0 false tags. Participant satisfaction ratings averaged 7.4 out of 10, indicating generally positive perceptions of the system despite usability challenges.

Participants with prior VR experience generally completed tasks more quickly and with fewer errors than participants without prior experience. For example, Dylan Daniel and Kaylee Moon, the two participants with previous VR experience, completed the task in under 4.5 minutes and reported relatively high satisfaction scores.

Participants without prior AR/VR experience often required additional time to adapt to the gesture controls and occasionally experienced accidental activations or failed gesture recognition. The slowest completion times were associated with higher numbers of placement errors and false tags, suggesting a relationship between gesture recognition difficulty and overall task efficiency.

6.2 Observed Trends

Several consistent usability trends were observed during the experiment:

- Users often required a short adjustment period before becoming comfortable with gesture interaction.
- False tag placements commonly occurred when gestures were performed too quickly or too close to unintended objects.
- Participants who completed tasks more quickly generally reported higher satisfaction scores.
- Left-handed participants occasionally reported difficulty performing gestures comfortably because the system was primarily optimized around right-handed interaction assumptions.
- Participants with prior VR experience adapted more rapidly to spatial interaction and gesture-based controls.

6.3 Participant Feedback

Participant feedback was generally positive regarding the overall concept and usefulness of the system. Several users described the interaction as intuitive and fun once they became familiar with the gestures. Participants appreciated the hands-free interaction design and the ability to place persistent markers within the environment.

However, participants also identified several usability concerns. The most common issue involved accidental tag placement caused by unintended gesture detection. Some participants also reported that the gesture controls occasionally felt inconsistent or difficult to perform precisely.

Multiple participants suggested adding:

- More customizable labels
- Voice labeling options
- Improved gesture calibration

7 Discussions

The results demonstrate that gesture based spatial labeling in augmented reality is both feasible and usable for novice users, although interaction reliability remains an important challenge. Most participants were able to complete the assigned tasks relatively successfully, and average satisfaction ratings shows generally positive impressions of the system.

One major finding was the relationship between prior VR experience and task performance. Participants with previous VR exposure adapted to the interface faster and completed tasks with fewer mistakes. This suggests that familiarity significantly affects user performance in AR systems.

The study also revealed that gesture based interaction can create usability issues related to accidental activations and gesture precision. False tag placements occurred frequently among participants unfamiliar with AR hand tracking systems. These findings align with prior research on mid-air interaction fatigue and gesture-based AR interfaces, which collectively suggest that natural interaction techniques often trade precision for convenience [28, 15].

Despite these issues, participants generally responded positively to the hands free interaction approach. Many users reported that the interface became more intuitive after a short learning period. This supports prior AR interaction research showing that gesture based interfaces can become more natural and efficient as users gain familiarity with the interaction style.[15, 23].

8 Limitations

This study has several limitations that should be considered when interpreting the results.

First, the sample size was limited to 12 participants, which restricts the generalizability of the findings. While consistent trends were observed, a larger and more diverse participant pool would be necessary to validate statistical significance and reduce potential sampling bias.

Second, all participants were recruited from a single area, Fort Collins, which may introduce a demographic homogeneity. This limits the ability to generalize findings to broader populations with different levels of technical literacy or AR familiarity.

Third, the system evaluated only gesture based interaction for labeling, without including a fully implemented voice input condition. As a result, this study does not directly measure the core comparison between voice and gesture under identical system conditions, but instead focuses on gesture usability.

Fourth, the experiment was conducted in a controlled environment rather than a real-world noisy or crowded setting. Prior research shows that AR interaction performance, especially for voice-based systems, is highly sensitive to environmental noise and context [16]. Therefore, the current results may overestimate performance in ideal conditions.

Finally, participants used a prototype system with predefined labels. This design choice may have restricted interaction patterns and influenced satisfaction ratings.

9 Future work

Future work should extend this study in several key directions.

First, a direct comparison between gesture based and voice based labeling within the same AR system should be implemented. This would allow controlled evaluation of speed, accuracy, and user preference under identical task conditions, especially under varying environmental noise levels as suggested in prior research [8].

Second, the system should be evaluated in more realistic environments, including noisy and crowded public spaces, to better understand how gesture and voice modalities perform under real-world constraints. This would align with findings that environmental context significantly impacts AR interaction performance [26].

Third, future work should include improved gesture recognition and calibration mechanisms to reduce unintended tag placements.

Finally, a broader demographic of participants would better diversify the results as our study's age range was from 20-23, severely limiting the feedback received to those of younger adults and a certain type of experience.

10 Conclusion

This project presented AR Place Tag, an augmented reality system designed to support spatial labeling of physical environments using gesture based interaction. The goal of this study was to evaluate the usability, efficiency, and user experience of marker placement and labeling in AR environments derived from gestures.

A within-subjects user study with 12 participants was conducted to measure task completion time, interaction errors, false tag placements, and satisfaction. Results showed that participants with prior VR experience adapted more quickly and achieved higher performance, while novice users required additional time to become familiar with the gesture system.

Overall, participants reported positive experiences with the system, particularly highlighting its hands-free and intuitive interaction design. However, issues related to gesture recognition accuracy and accidental tag placement were common, indicating areas for improvement in precision and feedback mechanisms.

The findings suggest that gesture based AR labeling is a promising approach for spatial labeling tasks, especially in contexts where voice input may be unreliable. However, improvements in accessibility and multimodal interaction support are necessary before such systems can be reliably used in real world environments.

Future work will focus on integrating voice input, improving gesture recognition reliability, and evaluating performance in noisy, real world conditions to better understand the trade-offs between input modalities in augmented reality systems.

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